

Unsupervised Learning and Dimensionality Reduction Report FAQ

CS7641: Machine Learning

Summer 2026

1 Frequently Asked Questions

This FAQ is intentionally extensive. Everything required for the assignment is already stated in the project description, but each term many of the same questions come up as students begin clustering, choosing component counts, interpreting dimensionality reduction, and writing the report. Use this FAQ as a guide while you work rather than trying to absorb it all at once.

The assignment is open-ended in analysis, but the expectations are concrete. The central idea is that you are not just running unsupervised algorithms. You are building a reproducible study of the Covertype feature space. Your job is to explain what structure exists before labels are used, whether that structure aligns with `Cover.Type`, how PCA/ICA/RP change the representation, and whether those lower-dimensional representations help or hurt clustering and downstream neural-network learning.

A strong report does not need every method to improve performance. A strong report needs controlled experiments, appropriate evidence, and careful interpretation.

2 Why This Assignment Feels Ambiguous

Q: Why does this assignment feel less straightforward than SL or OL?

Supervised learning assignments usually have a clear target and a clear evaluation loop. You train a model to predict a label, measure performance on held-out data, and compare algorithms using known outcomes. Unsupervised learning is messier. Clustering and dimensionality reduction do not know the labels during fitting, and there may not be a single “correct” answer.

This means you are not only asking whether a score is high. You are asking what kind of structure an algorithm found, whether that structure is stable, whether it is meaningful, whether it relates to known labels, and whether it helps downstream learning. Two methods can disagree and both can reveal something useful. A clustering solution can have good internal geometry but weak label alignment. A PCA representation can preserve most variance while losing information that matters for minority classes. An RP representation can work well on average but vary across random seeds. These are not nuisances to hide; they are often the main findings.

Q: What does “good analysis” mean when there is no single correct answer?

Good analysis means the report explains the relationship between method assumptions, data properties, and observed results. K-Means assumes a distance geometry where compact spherical clusters are meaningful. GMM/EM assumes mixture components that may be elliptical depending on covariance type. PCA preserves variance, not necessarily class information. ICA seeks non-Gaussian independent components, which can be unstable. RP approximately preserves distances under random projection, but results can vary by seed.

A strong report does not pretend these methods answer the same question. It explains what each method is looking for and what it found on Covertype.

Q: How should I handle ambiguous or conflicting results?

Conflicting results are expected. Internal clustering metrics may suggest one number of clusters, while label-alignment metrics suggest another. PCA may improve runtime but slightly reduce Macro-F1. ICA may produce high-kurtosis components that are difficult to interpret. RP may preserve neural-network performance but show seed sensitivity in clustering.

Do not force all metrics into a single winner. Explain the trade-off. If one method is better for clustering geometry, another for label alignment, and another for runtime, that is a valid and useful conclusion.

3 Project Scope

Q: What exactly is the task?

You will complete one unsupervised learning and dimensionality reduction study using the Forest Cover Type dataset, also known as Covertypes. The external label is `Cover_Type`, with classes 1 through 7. You should reuse the approximately 20,000-instance Covertypes sample from your SL Report unless a change is clearly justified.

You will run K-Means and EM/GMM on the original feature space, apply PCA, ICA, and Randomized Projections, rerun clustering after each dimensionality reduction method, and then rerun the SL PyTorch neural network using the best justified training recipe informed by SL and OL on each dimensionality-reduced representation.

Q: Is this a supervised clustering assignment?

No. The labels are available, but clustering and dimensionality reduction must be fit without using `Cover_Type`. Labels are used after fitting for interpretation, visualization, external clustering metrics, contingency tables, and downstream supervised neural-network evaluation.

This distinction matters. A clustering solution does not fail just because it does not perfectly match the labels. It may reveal a different kind of structure in the feature space, such as elevation, wilderness area, soil type, distance features, or mixed cartographic patterns.

Q: What algorithms are required?

The required clustering algorithms are **K-Means** and **Expectation Maximization via Gaussian Mixture Models**. The required dimensionality reduction methods are **PCA**, **ICA**, and **Randomized Projections**. For downstream supervised evaluation, you must compare the original-feature neural-network baseline against neural networks trained on PCA, ICA, and RP representations.

The extra credit option is one nonlinear manifold learning method, such as UMAP, t-SNE, Isomap, Locally Linear Embedding, or Spectral Embedding. Extra credit must add meaningful insight beyond the required linear methods.

Q: Is there a required “cluster-derived features” neural-network step?

No. The finalized Summer 2026 UL Report does not require training the neural network on cluster-derived features such as one-hot cluster IDs, centroid distances, or GMM responsibilities. The required downstream neural-network comparison is original features versus PCA, ICA, and RP features.

You may discuss cluster-derived features as a possible future direction, but do not treat them as a required section.

4 Dataset, Labels, and Preprocessing

Q: What dataset should I use?

Use the Forest Cover Type / Covertypes dataset from the SL Report. The working dataset should be approximately 20,000 instances sampled from the full dataset, matching the SL Report sample unless a change is

explicitly justified.

If you change the sample, preprocessing, split strategy, feature handling, or target mapping, you must disclose the change and explain how it affects comparison to prior SL and OL results.

Q: How should I think about the Covertypes features?

Covertypes contains continuous cartographic variables and binary wilderness-area / soil-type indicators. This mixed feature structure matters. Continuous features may have very different scales, while binary indicators carry categorical information as 0/1 variables.

Distance-sensitive methods such as K-Means are affected by scaling. PCA is affected by feature variance and covariance structure. ICA is affected by preprocessing, whitening, and non-Gaussianity. RP preserves geometry relative to the representation it is given. Neural-network training can also change depending on whether inputs are scaled or compressed.

Your report should explain whether you scaled all features, scaled only continuous features, left binary features unchanged, or used another defensible approach. There is not one universally correct choice, but there must be a rationale.

Q: What preprocessing choices should I think about before running anything?

Start with the geometry implied by your preprocessing. K-Means, GMM, PCA, ICA, and RP all operate on the feature matrix they receive, so preprocessing can change the problem the algorithms are solving.

For Covertypes, the major preprocessing decision is how to handle continuous cartographic features and binary indicator features together. Scaling continuous features is usually important because elevation and distance variables can dominate Euclidean distances and covariance structure. Binary wilderness and soil indicators may be left as 0/1, scaled with the rest of the matrix, or handled in a separate pipeline. Each choice changes the geometry. Scaling all features may give binary indicators larger relative influence than expected. Scaling only continuous features may preserve the meaning of binary indicators while still preventing large-scale continuous variables from dominating. Leaving everything unscaled may cause large-range features to drive K-Means, PCA, and distance-preservation checks.

A strong report does not need to test every preprocessing option, but it should explain the chosen one and connect that choice to algorithm behavior.

5 Label Use: Helpful, Dangerous, and Easy to Misuse

Q: Can I use Cover_Type labels during clustering or dimensionality reduction?

No. You may not use `Cover_Type` to fit K-Means, EM/GMM, PCA, ICA, RP, or a nonlinear manifold method. You also should not choose cluster counts or component counts solely by maximizing agreement with held-out labels.

You may use `Cover_Type` after fitting to interpret results. Appropriate uses include coloring a plot, computing ARI or NMI, building a contingency table, discussing which cover types appear mixed within clusters, and evaluating downstream supervised neural-network performance.

Q: Why are labels risky in an unsupervised assignment?

Labels are useful for interpretation, but they can quietly turn an unsupervised study into a supervised one. If you use labels to choose k , choose the number of GMM components, choose PCA dimensions, choose ICA dimensions, choose RP dimensions, or select the “best” unsupervised representation, then the unsupervised analysis is no longer label-free. It becomes an indirect supervised tuning process.

This is especially risky when labels are used from the test set. If you choose an unsupervised transformation because it gives the best test-label alignment or best test neural-network score, you have used the test set for model selection. That weakens the validity of your final conclusions.

Q: What if I do not use labels at all?

Not using labels at all is also a problem. The assignment asks whether unsupervised structure aligns with `Cover_Type`. You need labels after fitting to evaluate that relationship. Without external label evaluation, the report cannot answer whether clusters correspond to known cover types, whether minority classes are mixed into larger clusters, or whether dimensionality reduction preserves class-relevant structure.

The right approach is not “never use labels.” The right approach is **use labels only after unsupervised fitting**. Fit models without labels, then use labels as external evidence.

Q: What are appropriate ways to use labels?

Appropriate label use includes computing ARI, NMI, homogeneity, completeness, or V-measure after clustering is complete. It also includes building a cluster-label contingency table, coloring a fitted embedding by `Cover_Type`, discussing which classes are mixed or separated, and evaluating downstream supervised neural-network performance after a leakage-safe train/validation/test split.

Labels can help you interpret what the algorithms found. They should not determine what the algorithms are allowed to find.

Q: What are inappropriate ways to use labels?

Inappropriate label use includes fitting clustering or dimensionality reduction with `Cover_Type`, picking k because it maximizes test-set ARI, choosing PCA dimensions because they maximize test accuracy, selecting the best RP seed because it gives the best test Macro-F1, or fitting transformations on the full dataset before downstream neural-network evaluation.

A useful rule: labels may explain a result after fitting, but they should not control the fitting process.

Q: Do I need train, validation, and test splits for unsupervised learning?

For exploratory unsupervised analysis, you may fit clustering and dimensionality reduction on the working sample as long as labels are only used after fitting for interpretation. The goal there is to understand the structure of the feature space.

For downstream neural-network evaluation, you must be stricter. Any learned preprocessing or dimensionality reduction transformation used before the neural network must be fit on the training data only and then applied to validation and test data. This includes scaling, PCA, ICA, RP, and any other learned transformation. Fitting PCA or scaling on the full dataset before supervised evaluation is leakage.

Q: What is the most common leakage mistake in this assignment?

The most common leakage mistake is fitting dimensionality reduction on the full dataset before evaluating neural-network performance. For example, fitting PCA on all 20,000 examples and then splitting into train/test gives the test set influence over the learned representation. That invalidates the supervised downstream comparison.

The correct downstream workflow is: split first, fit scaling and DR on training data, transform validation and test using the fitted transformation, train the NN on transformed training data, tune using validation data, and evaluate once on the held-out test set.

6 Hypotheses and Study Design

Q: How many hypotheses do I need?

You need at least **one clear, testable hypothesis** before presenting the main results. You may include more than one if it helps your narrative, but one strong hypothesis is enough.

The hypothesis may focus on clustering, dimensionality reduction, label alignment, downstream neural-network behavior, or a relationship among these pieces.

Q: What makes a hypothesis strong?

A strong hypothesis predicts a behavior or trade-off and ties it to dataset properties or course concepts. It should be something your experiments can actually support, contradict, or qualify.

A weak hypothesis says, “PCA will perform best.” A stronger hypothesis says, “PCA will preserve downstream neural-network performance better than ICA or RP because Covertypes contains correlated continuous cartographic features, so the leading principal components may retain most broad geometric structure while reducing input dimension.” That version predicts both behavior and mechanism.

Another strong hypothesis might say, “K-Means will only partially align with `Cover_Type` because Euclidean geometry may capture elevation and wilderness-area patterns more directly than ecological cover labels.” This sets up a meaningful cluster-label alignment analysis.

Q: Where should I revisit the hypothesis?

Revisit the hypothesis in the Discussion or Conclusion. Do not just restate it. Use evidence from clustering metrics, label-alignment metrics, DR diagnostics, DR + clustering results, neural-network performance, runtime, or class-level behavior to say whether the hypothesis was supported, contradicted, or qualified.

7 Clustering on Original Features

Q: What should I do for K-Means?

Run K-Means on the original Covertypes feature space using a justified preprocessing pipeline. You should explore at least three reasonable values of k , report initialization and restart settings such as `n_init`, fix or report random seeds, and justify the final selected value or comparison range.

The value $k = 7$ is a useful reference because there are seven cover-type labels, but it is not automatically the correct number of clusters. K-Means is unsupervised. The natural feature geometry may support fewer or more clusters than the number of labels.

Q: What K-Means hyperparameters should I think about?

The most important K-Means hyperparameter is k , the number of clusters. Small k values force broad groupings and may merge several cover types or environmental regimes. Large k values can split meaningful groups into smaller subclusters, sometimes revealing structure that does not map directly to labels. Since Covertypes has seven labels, $k = 7$ is a useful reference, but a good sweep should include values below and above seven.

The initialization strategy also matters. K-Means can converge to different local optima depending on starting centroids. Using `k-means++` and reporting `n_init` helps reduce sensitivity to unlucky starts. If different seeds produce noticeably different inertia, silhouette, or label alignment, that instability is worth discussing.

The distance geometry is another hidden hyperparameter. K-Means uses Euclidean geometry by default, so scaling decisions determine which features dominate cluster assignment. If elevation or distance features dominate the clusters, the report should connect that to preprocessing and feature scale.

Q: What should I expect K-Means to do on Covertypes?

K-Means may find broad cartographic groupings rather than cover-type labels. It may separate examples by elevation bands, distance-to-road or distance-to-hydrology structure, wilderness areas, or soil-type indicator patterns. This is not necessarily bad. It means the strongest Euclidean structure in the feature space may not be the same as the supervised label structure.

A common outcome is moderate internal structure but weak label alignment. If that happens, do not stop at “K-Means failed.” Ask what the clusters contain. Compare cluster feature summaries, cluster-label contingency tables, and internal metrics.

Q: What should I do for EM/GMM?

Use Gaussian Mixture Models as the EM-based clustering method. You should explore at least three reasonable component counts and report the covariance type, initialization, convergence criteria, convergence behavior, and random seed handling.

The covariance type matters. A full covariance GMM is more flexible but may be more expensive or less stable. A diagonal, tied, or spherical covariance assumption may be more constrained but easier to interpret. The report should explain why the selected covariance structure is appropriate for the data and sample size.

Q: What GMM/EM hyperparameters should I think about?

The number of components is the GMM analogue of k , but GMM components are probabilistic mixture components rather than hard centroid clusters. A component count of seven is a reasonable reference, not a requirement. Fewer components may capture broad regimes. More components may model substructure within cover types or environmental regions.

Covariance type changes the shape of the mixture components. `spherical` assumes each component has one variance value and is closest to K-Means-like geometry. `diag` allows each feature to have its own variance within each component but no covariance between features. `tied` shares one covariance matrix across components. `full` allows each component to have its own full covariance matrix, which is flexible but more prone to instability and overfitting.

Initialization and regularization also matter. `n_init`, `random_state`, `max_iter`, `tol`, and covariance regularization such as `reg_cvar` in scikit-learn can affect convergence. If a GMM fails to converge or produces degenerate components, that is not just a software problem; it is evidence that the model assumptions or settings may not fit the data well.

Q: What should I expect GMM/EM to do on Covertypes?

GMM may outperform K-Means on likelihood or AIC/BIC while not improving label alignment. That can happen because GMM is modeling density structure, not labels. A flexible covariance model may capture elliptical feature distributions that K-Means cannot, but it may also fit statistical structure that is not class-discriminative.

Soft responsibilities can be useful. If many points have uncertain membership across several components, the data may not naturally form clean hard clusters. This is a meaningful contrast with K-Means and should be discussed.

Q: Which clustering metrics should I use?

Use at least two internal or model-selection metrics for original-space clustering. Examples include silhouette score, Davies–Bouldin index, Calinski–Harabasz index, inertia or elbow analysis for K-Means, and log-likelihood, AIC, or BIC for GMM.

You should also use at least one external label-alignment metric or summary after clustering is complete. Examples include adjusted Rand index, normalized mutual information, homogeneity, completeness, V-measure, or a cluster-label contingency table.

Internal metrics and external label-alignment metrics answer different questions. Internal metrics ask whether clusters are compact, separated, or statistically plausible in the feature space. External metrics ask whether those clusters align with `Cover_Type`. These may disagree, and that disagreement can be one of the most interesting findings in the report.

Q: How should I interpret common clustering metrics?

Silhouette score compares how close points are to their assigned cluster relative to other clusters. Higher is generally better, but it can favor compact, separated, spherical clusters and may not reward complex structure.

Davies–Bouldin index compares within-cluster scatter to between-cluster separation. Lower is generally better. It can be useful for comparing settings, but it still assumes that compactness and separation are the right goals.

Calinski–Harabasz index compares between-cluster dispersion to within-cluster dispersion. Higher is generally better. It can sometimes favor larger numbers of clusters, so it should not be used alone.

Inertia is the K-Means within-cluster sum of squares. It decreases as k increases, so the absolute minimum is not meaningful by itself. Look for elbows, diminishing returns, and whether the resulting clusters are interpretable.

For GMM, log-likelihood usually improves as model flexibility increases. AIC and BIC penalize model complexity and are often more useful for comparing component counts or covariance types. BIC penalizes complexity more strongly than AIC. A model with better BIC does not necessarily have better label alignment.

ARI and NMI evaluate agreement with labels after clustering. ARI adjusts for chance and is often strict. NMI measures shared information between clusters and labels and can be easier to interpret across cluster counts. Homogeneity, completeness, and V-measure separate whether clusters contain mostly one class and whether class members are grouped together. Use these as external interpretation, not as fitting objectives.

Q: What if my clusters do not match Cover_Type?

That is not automatically bad. The feature space may contain structure that is not identical to the class labels. For example, clustering may organize examples by elevation, distance-to-hydrology, wilderness-area indicators, or soil-type groups rather than by cover type.

A weak report says, “The ARI was low, so clustering failed.” A stronger report says, “The ARI was low, but the contingency table and feature summaries suggest that K-Means separated broad elevation and wilderness-area structure rather than cover-type labels. This indicates that the natural Euclidean geometry of the feature space differs from the supervised label structure.”

8 Dimensionality Reduction

Q: What is the purpose of dimensionality reduction in this assignment?

Dimensionality reduction is not just a way to make pretty plots. It should help you reason about representation quality. You are asking whether PCA, ICA, and RP preserve useful structure, remove noise, distort distances, change cluster behavior, or alter downstream neural-network performance.

A good dimensionality reduction section explains what each method preserves and what it may discard.

Q: What should I report for PCA?

For PCA, report explained variance ratios, cumulative explained variance, and a justified component count. A scree plot or cumulative variance curve is usually useful. You may also report reconstruction error, effective rank, or component loadings if they support your interpretation.

Do not assume that retaining a large percentage of variance means retaining all class-relevant information. High-variance directions may capture broad cartographic structure, while lower-variance directions may still matter for minority classes or class boundaries.

Q: What PCA hyperparameters and settings should I think about?

The main PCA setting is `n_components`. A small number of components creates strong compression and may remove noise, but it can also discard class-relevant information. A larger number preserves more variance but may offer less compression and less runtime benefit.

The scaling choice before PCA is critical. PCA follows variance. If features are not scaled, large-range variables can dominate the first principal components. If all features are standardized, binary indicators and continuous features are placed on a more comparable variance scale. The report should explain why the chosen scaling matches the intended analysis.

Solver settings are usually less central to interpretation, but they affect reproducibility and runtime. In scikit-learn, `svd_solver` may be `auto`, `full`, `randomized`, or another supported option depending on the setup. If a randomized solver is used, report the seed. Whitening is another important option. PCA whitening rescales components to unit variance, which can help some downstream methods but removes the original variance scale. If whitening is used, justify it.

Q: How should I interpret PCA explained variance?

The explained variance ratio for a component tells you what fraction of total variance is captured by that component. The cumulative explained variance tells you how much total variance is captured by the first m components.

A steep cumulative curve suggests redundancy or low-dimensional structure. A slow curve suggests that variance is spread across many directions. A visible elbow suggests a possible component-count choice, but the elbow is not automatically optimal for clustering or neural-network performance.

Explained variance is not the same as class information. A component can explain a lot of variance because it captures broad elevation or distance structure, while a lower-variance component might still help distinguish a minority cover type. This is why PCA diagnostics should be connected to clustering, label alignment, and downstream NN behavior.

Q: What is PCA reconstruction error, and when is it useful?

PCA reconstruction error measures how well the reduced representation can reconstruct the original feature matrix. Lower reconstruction error means less information was lost in a squared-error sense. It can help justify component count, especially when comparing strong compression against moderate compression.

Reconstruction error is still not a supervised metric. A representation can reconstruct the feature matrix well while losing subtle class-discriminative information. Use reconstruction error as one piece of evidence, not the whole argument.

Q: What are PCA loadings, and should I use them?

PCA loadings describe how original features contribute to each principal component. They can help interpret what the component represents. For example, a component may be dominated by elevation and distance features, or by groups of binary soil indicators.

You do not need to list every loading. A compact table of the largest positive and negative loadings for the first few components can help explain whether PCA is capturing meaningful domain structure. This is especially useful if your clustering or NN results change after PCA.

Q: What should I report for ICA?

For ICA, report a justified component count, non-Gaussianity evidence such as kurtosis, convergence behavior, and some discussion of seed sensitivity or initialization. ICA can be unstable, and that instability is worth discussing rather than hiding.

ICA is not just PCA with a different name. PCA seeks orthogonal directions of maximum variance. ICA seeks statistically independent, non-Gaussian components. Your report should explain whether the ICA components appear meaningful, unstable, sparse, or tied to particular feature groups.

Q: What ICA hyperparameters and settings should I think about?

The main ICA setting is `n.components`. Too few components may miss independent structure. Too many components may increase instability, convergence problems, or noisy components. Component count should be justified using non-Gaussianity evidence, stability, convergence behavior, or downstream utility.

Whitening is central to ICA. FastICA typically whitens the data before estimating independent components. The whitening choice and preprocessing scale affect the components you find. If you use scikit-learn, report the relevant whitening behavior, component count, random seed, maximum iterations, tolerance, and whether the algorithm converged.

The nonlinearity or contrast function also matters. In FastICA, options such as `logcosh`, `exp`, or `cube` correspond to different approximations for non-Gaussianity. You do not need a huge sweep, but if you change the nonlinearity, explain why. Random seed matters because ICA can converge to different component orderings, signs, or solutions.

Q: How should I interpret ICA kurtosis?

Kurtosis measures tail-heaviness or non-Gaussianity relative to a Gaussian distribution. ICA seeks components that are as statistically independent and non-Gaussian as possible, so component kurtosis can help evaluate whether ICA found non-Gaussian structure.

High absolute kurtosis may indicate a component with sparse, heavy-tailed, or outlier-driven behavior. Low kurtosis may suggest a component closer to Gaussian. However, high kurtosis is not automatically useful. It may reflect meaningful sparse structure, or it may reflect noise, rare binary patterns, or outliers. The report should connect kurtosis to component interpretation and downstream results.

Use absolute kurtosis or excess kurtosis consistently and state which one you use. A compact table showing the distribution of component kurtosis across component counts can be more useful than a long list of component values.

Q: What does ICA convergence tell me?

Convergence tells you whether the optimization procedure stabilized under the chosen settings. If ICA reaches `max_iter` without converging, that is important. It may indicate that the component count is too high, preprocessing is problematic, the tolerance is too strict, or the data does not fit ICA assumptions cleanly.

Do not hide convergence warnings. They are part of the analysis. A method that produces unstable components or fails to converge may be less trustworthy even if one downstream score looks good.

Q: How do I evaluate ICA stability?

ICA components can change order and sign across seeds, so comparing components directly can be tricky. A sign flip does not necessarily mean the component changed meaning. Component order may also permute.

Useful stability checks include comparing distributions of kurtosis across seeds, comparing downstream clustering or NN results across seeds, or matching components across runs using absolute correlations. The goal is not perfect component identity. The goal is to know whether the ICA representation is stable enough to support your claims.

Q: What should I report for Randomized Projections?

For Randomized Projections, report a justified component count and evaluate variability across multiple random seeds. Since RP is stochastic, one projection is not enough if you are making a claim about its behavior.

You should report distance preservation, projection distortion, or reconstruction only if you implement a valid inverse or pseudo-inverse reconstruction. RP is not primarily a reconstruction method. If you do not evaluate reconstruction, focus on whether pairwise distances, clustering behavior, and downstream neural-network performance are preserved across seeds.

Q: What RP hyperparameters and settings should I think about?

The main RP setting is the number of components. Fewer components give stronger compression and faster downstream models, but they may distort distances more. More components should preserve geometry better but reduce the benefit of dimensionality reduction.

The projection type also matters. Gaussian Random Projection uses a dense random matrix. Sparse Random Projection can be faster and more memory efficient. Both are valid if justified. Since RP is random, report seeds and evaluate variability. If one seed performs unusually well, do not treat that as the method's general behavior without additional evidence.

Q: How do I choose component counts?

Choose component counts using method-appropriate evidence. PCA may use cumulative explained variance, scree plots, reconstruction error, or effective rank. ICA may use non-Gaussianity, convergence behavior, stability, and downstream usefulness. RP may use distortion, distance preservation, runtime, and seed variability.

You do not need to use the same number of components for PCA, ICA, and RP. If you use different component counts, explain why. If you use the same component count for all methods, explain why that comparison is meaningful.

Q: What does it mean to compare representations across PCA, ICA, and RP?

A good comparison does not simply report three final scores. It explains what each representation appears to preserve. PCA may preserve dominant variance, ICA may emphasize non-Gaussian structure, and RP may preserve distances approximately while introducing randomness.

A compact comparison table can be very effective. Useful columns might include component count, main diagnostic, stability, runtime, clustering result, label alignment, and downstream NN result.

9 Clustering after Dimensionality Reduction

Q: What exactly is required for DR + clustering?

You must rerun both clustering algorithms after each required dimensionality reduction method. That means K-Means after PCA, K-Means after ICA, K-Means after RP, GMM after PCA, GMM after ICA, and GMM after RP.

You do not need to give equal prose space to all six combinations, but you must report enough evidence to compare them fairly.

Q: Do I need to reselect k or GMM components after DR?

Yes, or you must justify why you are intentionally reusing the original-space settings. Dimensionality reduction changes the feature geometry, so the best clustering settings may also change.

For K-Means, evaluate or defend k in the transformed space. For GMM, evaluate or defend the number of components and covariance type in the transformed space. Reusing the same settings everywhere is acceptable only if the report explains why that comparison is meaningful.

Q: What should I compare in the DR + clustering section?

Compare original-space clustering to clustering after PCA, ICA, and RP. Use internal clustering metrics and external label-alignment evidence. Then explain what changed.

The key question is not only which combination had the best score. The better question is why the representation changed clustering behavior. PCA might filter noise or remove class-relevant low-variance structure. ICA might reveal non-Gaussian patterns or introduce instability. RP might preserve geometry well at moderate dimensions or produce seed-dependent clustering.

Q: What should I expect after DR is applied before clustering?

PCA may improve clustering if it removes noisy or redundant directions, but it may hurt label alignment if lower-variance class-relevant information is discarded. ICA may reveal sparse or non-Gaussian structure, but it may also be unstable or harder for K-Means/GMM to use cleanly. RP may preserve distance structure surprisingly well, but results may vary across seeds.

The important question is whether the transformed space changes the geometry in a way that helps the clustering objective. Improved internal metrics do not guarantee improved label alignment. Improved label alignment does not guarantee more meaningful unsupervised geometry. Discuss both.

Q: How should I present this section without using too much space?

A compact table is usually best. Rows can represent original space, PCA, ICA, and RP. Columns can represent K-Means internal metric, K-Means label alignment, GMM internal/model-selection metric, GMM label alignment, and a short note.

You can place extended tables in the REPRO PDF, but the main report must still include enough evidence and interpretation to stand alone.

10 Neural Network after Dimensionality Reduction

Q: Which neural network should I use?

Use the SL PyTorch neural network with the best justified training recipe informed by your SL and OL Reports. This may reflect your SL architecture choice, OL optimizer findings, OL regularization findings, or other prior evidence.

This section is not a new architecture search. Since the input dimension changes after PCA, ICA, or RP, you may adjust the input layer. Other changes to architecture, optimizer, regularization, training budget, or evaluation protocol should be minimized and justified.

Q: What comparisons are required?

Compare the original-feature neural-network baseline to neural networks trained on PCA, ICA, and RP representations. Use Accuracy, Macro-F1, balanced accuracy, class-level behavior, and runtime or another justified efficiency measure.

The comparison should use the same train/validation/test split and comparable training budgets unless a change is explicitly justified. The purpose is to isolate the effect of the representation, not to give each representation a completely different training setup.

Q: How do I avoid leakage in NN-after-DR?

Split first. Fit preprocessing and dimensionality reduction only on the training data. Apply the fitted transformations to validation and test data. Then train and evaluate the neural network.

This applies to PCA, ICA, RP, scaling, and any other learned transformation. If you use cross-validation or multiple validation splits, fit transformations inside the appropriate training fold or training split.

Q: What if DR makes neural-network performance worse?

That can be a meaningful result. Dimensionality reduction may remove information useful for class boundaries or minority classes. It may preserve broad structure while discarding subtle class-relevant features.

A good analysis explains whether the performance change appears to come from information loss, denoising, compression, altered geometry, faster training, instability, or class-level effects. Small score changes should not be overinterpreted unless supported by class-level behavior, runtime, repeated seeds, or consistent validation/test patterns.

Q: What should I expect from the NN-after-DR experiments?

PCA may preserve performance with fewer inputs if the original features contain redundancy. It may also reduce runtime or stabilize training. ICA may help if non-Gaussian components capture useful structure, but it may hurt if the representation is unstable or poorly aligned with class boundaries. RP may preserve performance surprisingly well if enough dimensions are retained, but it should be judged across seeds.

The original-feature NN is the baseline. A DR representation can be useful even if it does not beat the baseline, especially if it greatly reduces runtime or preserves performance with fewer features. It can also be harmful if it reduces Macro-F1, balanced accuracy, or minority-class recall.

11 Extra Credit: Nonlinear Manifold Learning

Q: What methods can I use for extra credit?

You may use one nonlinear manifold learning method such as UMAP, t-SNE, Isomap, Locally Linear Embedding, or Spectral Embedding. You must justify the choice and explain what it contributes beyond PCA, ICA, and RP.

Q: Can I use t-SNE or UMAP for downstream neural-network features?

Be careful. Some nonlinear methods, especially t-SNE, are primarily visualization tools and do not naturally support standard train/test transformation in the same way PCA does. If you use a nonlinear method for downstream neural-network features, you must explain how train, validation, and test transformations are handled without leakage.

It is acceptable to use nonlinear manifold learning only for visualization and interpretation, as long as the result adds meaningful insight.

Q: What earns strong extra credit?

Strong extra credit includes a justified method choice, clear hyperparameters, a visualization or quantitative diagnostic, comparison to at least one required linear method, and interpretation of whether the nonlinear method reveals structure that PCA, ICA, and RP did not.

Simply showing a colorful 2D plot without analysis is not enough for full extra credit.

12 Figures, Tables, and Page Planning

Q: What figures and tables should I include?

The exact layout is up to you, but the main report should include compact evidence for each major claim. Useful items include an original-space clustering metric table, cluster-label contingency table, PCA explained variance curve, ICA kurtosis or non-Gaussianity summary, RP distance-preservation or seed-variability summary, DR + clustering comparison table, and NN-after-DR metric/runtime table.

You do not need to show every plot you generated. The main report should show the evidence needed to support the story. Extended diagnostics can go in the REPRO PDF.

Q: How do I fit everything into 8 pages?

Use tables and multi-panel figures carefully. A single compact table can cover many clustering combinations. A small number of clear, interpreted figures is better than many tiny plots with unreadable labels.

You should avoid using page space to restate legends or describe obvious visual details. Instead, explain what the figure means. For example, do not only say that the PCA curve rises quickly. Explain whether that suggests redundancy, compression potential, or possible loss of minority-class information.

Q: How readable do figures need to be?

Figures and tables must be legible at normal grading zoom. Axis labels, legends, captions, and table text should be readable without excessive zooming. This is a scored requirement in the rubric.

If you need to shrink a figure until it becomes unreadable, it should probably be moved to the REPRO PDF or replaced with a clearer summary.

13 Discussion and Writing

Q: What should the Discussion do?

The Discussion should synthesize the full study. It should not simply repeat K-Means results, then GMM results, then PCA results, then NN results. It should explain what the results reveal about the Covertypes feature space.

A strong Discussion explains whether meaningful unsupervised structure exists, whether that structure aligns with `Cover_Type`, how K-Means and GMM differ, how PCA/ICA/RP change the representation, whether DR improves clustering or downstream learning, and what limitations affect the conclusions.

Q: What does cluster-label alignment mean conceptually?

Cluster-label alignment asks whether unsupervised groups correspond to known cover-type labels. Weak alignment does not automatically mean poor clustering. It may mean the labels do not match the strongest geometric structure in the feature space.

A useful Discussion should explain whether clusters align with some classes more than others, whether minority classes are mixed into larger clusters, and whether the clustering appears to capture feature structure that differs from ecological cover labels.

Q: What limitations should I discuss?

Limitations should be specific to the study. Useful limitations include the approximately 20,000-instance sample, class imbalance, mixed continuous/binary feature types, sensitivity to scaling, ICA convergence instability, RP seed variability, disagreement between internal and external metrics, or the limits of using class labels to judge unsupervised structure.

A weak limitation is “more work could be done.” A stronger limitation is “RP results varied across seeds, so the observed improvement in K-Means label alignment may not be stable without additional projections or a larger component count.”

Q: How do I avoid writing a checklist?

A checklist says what you ran. A technical paper explains why the results happened and what they mean.

Weak:

K-Means had low ARI. PCA kept 95% variance. The NN was faster after PCA.

Stronger:

K-Means produced weak label alignment despite moderate internal separation, suggesting that the dominant Euclidean structure of Covertypes does not map directly onto the seven cover-type labels. PCA preserved most downstream NN performance while reducing runtime, but the class-level metrics suggest that compression may have removed information relevant to minority classes.

Use the stronger style. Connect results to evidence and method assumptions.

14 Reproducibility and Submission

Q: What should go in the REPRO PDF?

The REPRO PDF should include the read-only Overleaf link, GitHub commit hash, exact run instructions, environment details, package versions, data paths, random seeds, sample confirmation, preprocessing details, clustering and DR hyperparameters, component counts, metric calculations, and extended diagnostics that do not fit in the main report.

The REPRO PDF supports reproducibility, but it does not replace the main report. The main report still needs to include the central evidence and interpretation.

Q: Do I need to use the IEEE conference template?

Yes. The main report must use the IEEE conference template. The report should use a format such as:

```
\documentclass[conference]{IEEEtran}
```

A generic article template or custom report format is not acceptable for the main report.

Q: Do I need to use Georgia Tech / Enterprise Overleaf?

Yes. The main report must be created using your Georgia Tech / Enterprise Overleaf account. You must provide a read-only Overleaf link. Do not send email invitations.

Q: Do I need to use GT Enterprise GitHub?

Yes. Course-related code must be stored in GT Enterprise GitHub. Your reproducibility sheet must include a commit hash from the final version of your code. Personal GitHub repositories are not acceptable for this assignment.

15 Sources, AI Use, and Academic Integrity

Q: Do I need outside sources?

Yes. You need at least two peer-reviewed references beyond course materials. Use sources to justify or interpret choices related to clustering validation, K-Means, Gaussian mixtures and EM, PCA, ICA, randomized projections, manifold learning, label-alignment metrics, or dimensionality reduction effects.

Primary sources are strongly preferred. Textbooks can be useful for background, but they should generally be avoided for the required outside-source expectation because they are too broad for supporting specific experimental choices. Avoid counting arXiv papers, preprint servers, blogs, documentation pages, tutorials, lecture notes, Wikipedia, Medium posts, or vendor documentation as one of the required peer-reviewed references.

Course blog posts and documentation can still be useful learning resources, but they do not replace the required peer-reviewed sources.

Q: Can I use AI tools?

Yes, with disclosure. You may use AI tools for brainstorming, outlining, grammar edits, code generation, code refactoring, debugging, and figure-code help. You may not submit AI-written analysis or conclusions as your own.

Every submission must include an AI Use Statement identifying the tools used, what they helped with, and confirming that you reviewed and understood all assisted content.

Q: Can I reuse code from libraries or examples?

General-purpose library code is allowed. Code, notebooks, templates, or automated exploration scripts written specifically for CS7641 assignments are not allowed. Your analysis, figures, and written interpretation must be original and grounded in your own experiments.

16 Quick Self-Check

Use this table before submitting. It is not the full rubric, but it catches most common problems.

Check	What to confirm
Dataset	I used Covertypes and treated <code>Cover_Type</code> classes 1 through 7 as external labels for evaluation and interpretation.
Sample	I reused the approximately 20,000-instance SL sample or clearly justified any change.
Preprocessing	I explained scaling and preprocessing choices for mixed continuous and binary features.
Label discipline	I did not use labels to fit clustering or dimensionality reduction.
Label evidence	I used labels after fitting to evaluate alignment, interpret clusters, or assess downstream supervised performance.
Hypothesis	I stated at least one clear hypothesis before results and revisited it with evidence.
K-Means	I tested multiple reasonable k values and reported initialization, seeds, metrics, and interpretation.
GMM/EM	I tested multiple component counts and reported covariance type, convergence behavior, seeds, metrics, and interpretation.
PCA	I reported explained variance, cumulative variance or reconstruction evidence, loadings or structure where useful, and a justified component count.
ICA	I reported non-Gaussianity evidence, kurtosis, convergence behavior, component-count rationale, and stability or seed considerations.
RP	I reported component-count rationale, seed variability, and distance preservation, distortion, or valid reconstruction evidence.
DR + clustering	I reran K-Means and GMM/EM after PCA, ICA, and RP and compared original-space and transformed-space clustering.
NN after DR	I compared original-feature NN performance against PCA, ICA, and RP features using consistent splits, metrics, and runtime evidence.
Leakage controls	I fit DR and preprocessing on training data only for downstream NN evaluation.
Figures and tables	My figures and tables are legible at normal grading zoom and each has an interpretive takeaway.
Discussion	I synthesized unsupervised structure, label alignment, method assumptions, DR effects, NN behavior, limitations, and practical takeaways.
Format	I used the IEEE conference template for the main report.
Overleaf	I used Georgia Tech / Enterprise Overleaf and included a read-only link.
GitHub	I used GT Enterprise GitHub and included a final commit hash.
Sources	I cited at least two peer-reviewed sources in IEEE style and used them in the body of the report.
AI statement	I included an AI Use Statement.
REPRO	I submitted a REPRO PDF with commands, seeds, environment details, sample confirmation, hyperparameters, and extended diagnostics.

17 Version Control

- v1.0 – 05/19/2026 – TJL refactored UL FAQ for Summer 2026 Covertypes assignment.

FAQ document written by Theodore LaGrow.